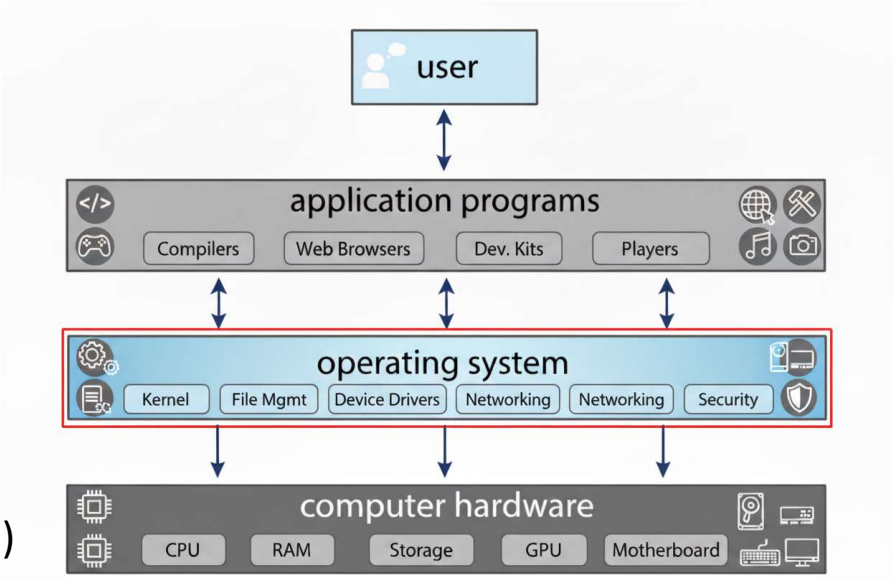
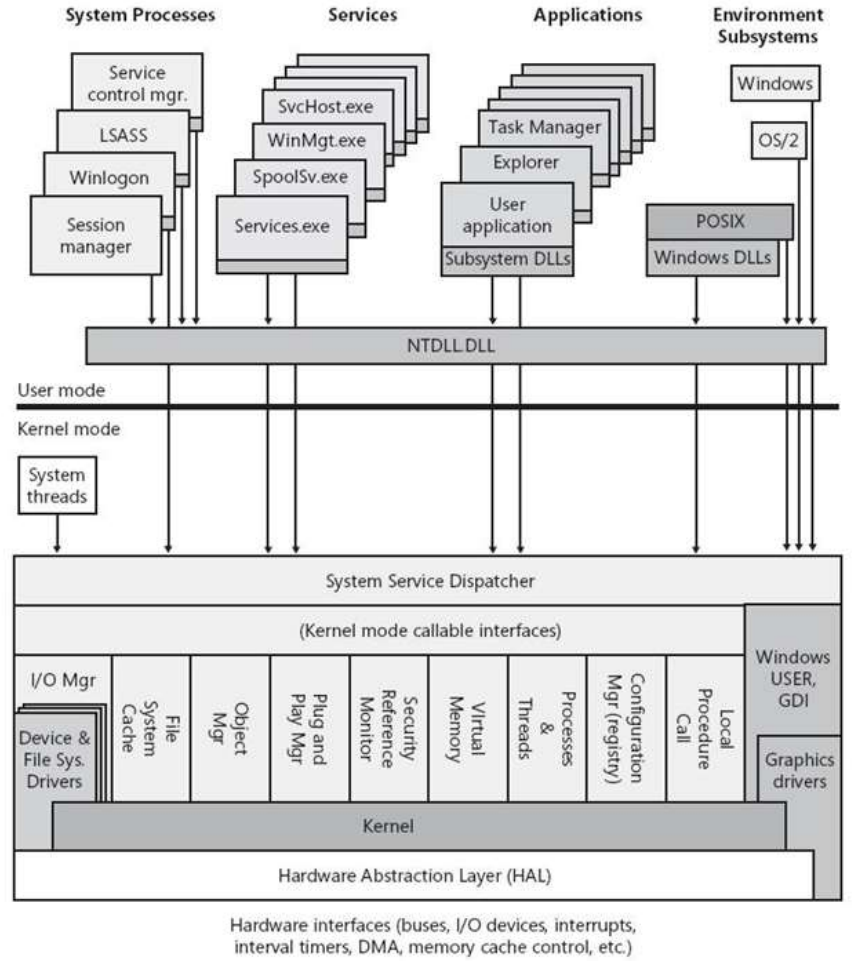
Introduction

What Operating Systems Do

- An operating system (OS) is software that manages computer hardware and provides a basis for application programs
  - To provide a convenient and efficient environment for executing programs
- **User View:** Varies by device (PC, mobile, embedded)
  - For PCs, the goal is ease of use and performance
  - For embedded systems, the goal is to run without user intervention
- **System View:**
  - The OS acts as a resource allocator (managing CPU, memory, storage)
  - And as the control program (preventing errors and improper use)
- **Kernel:** The core of the OS, the program that is running at all times
  - System programs – associated with the OS but not necessarily part of the kernel
  - Middleware – framework that provide additional services for application developers



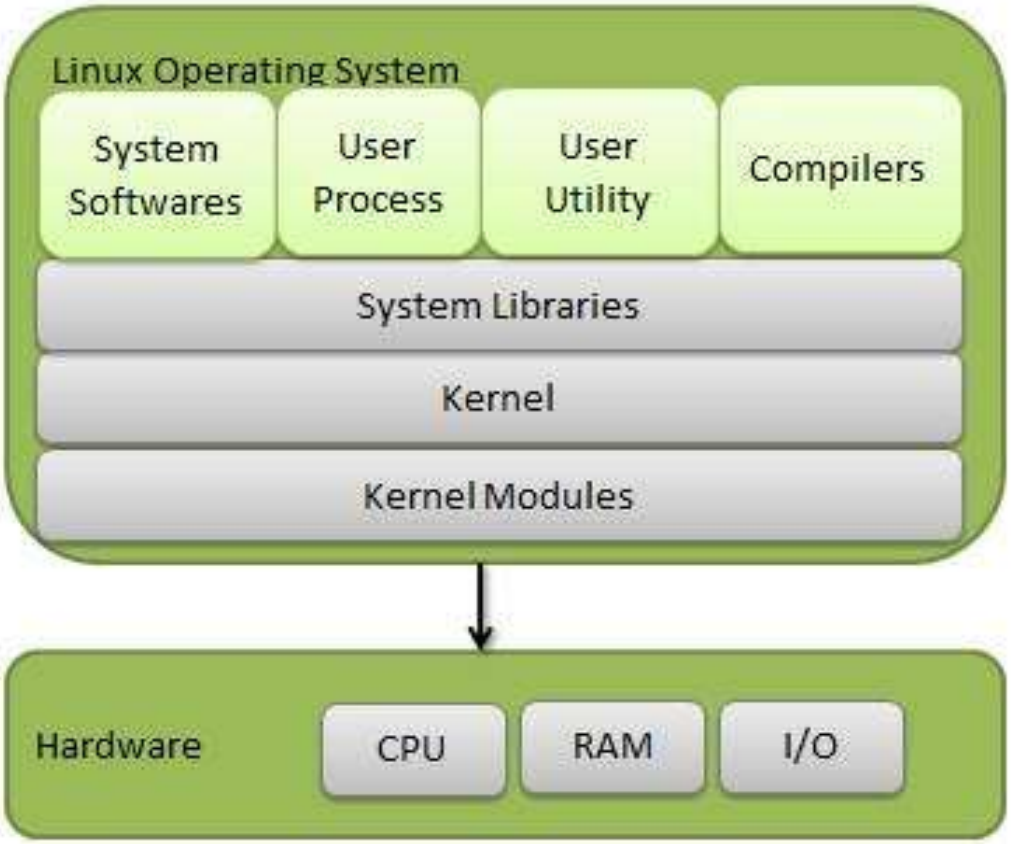
Windows Operating System Components



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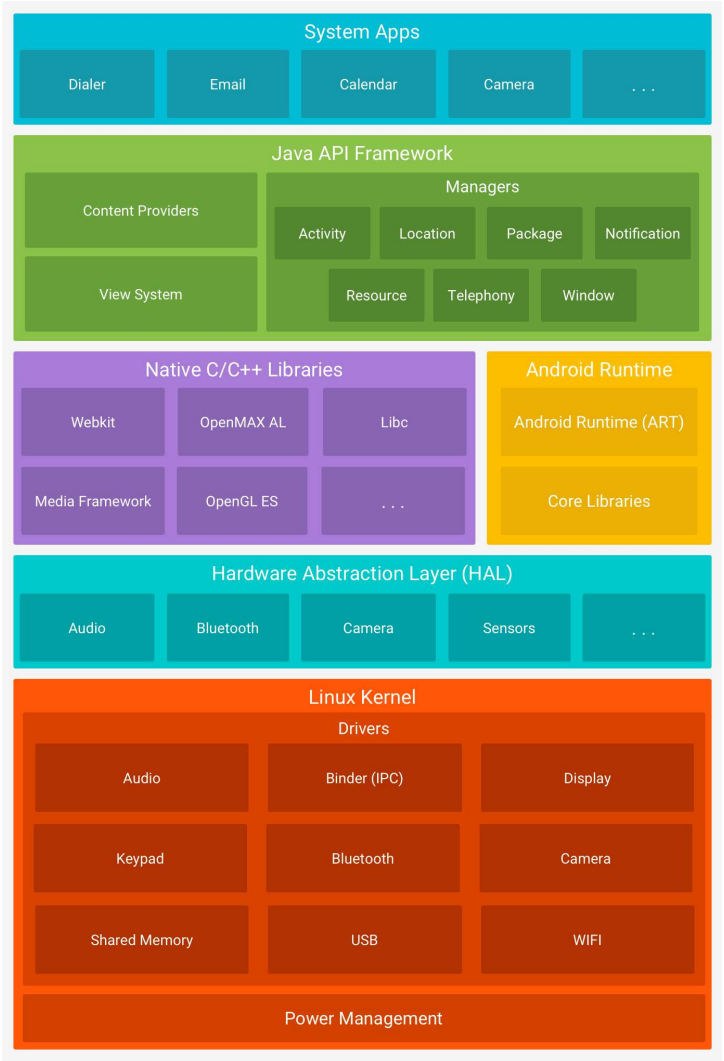
Retrieved from: https://learn.microsoft.com/en-us/windows-hardware/drivers/kernel/overview-of-windows-components

Linux Operating System Structure



Retrieved from: https://www.tutorialspoint.com/operating_system/os_linux.htm

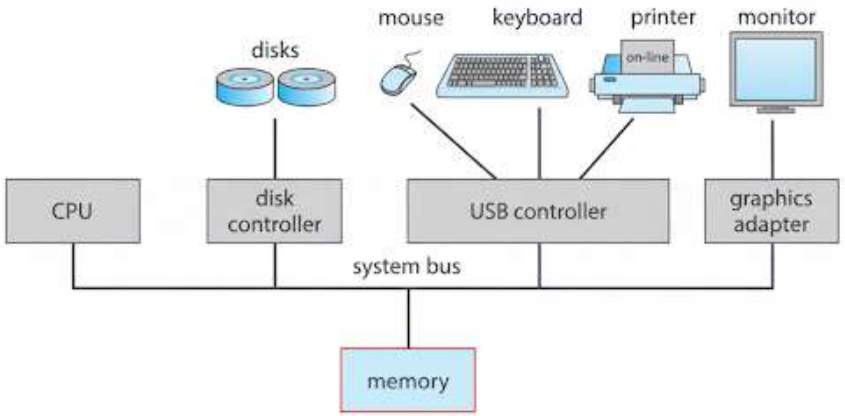
Android Operating System Structure



Retrieved from: https://developer.android.com/guide/platform

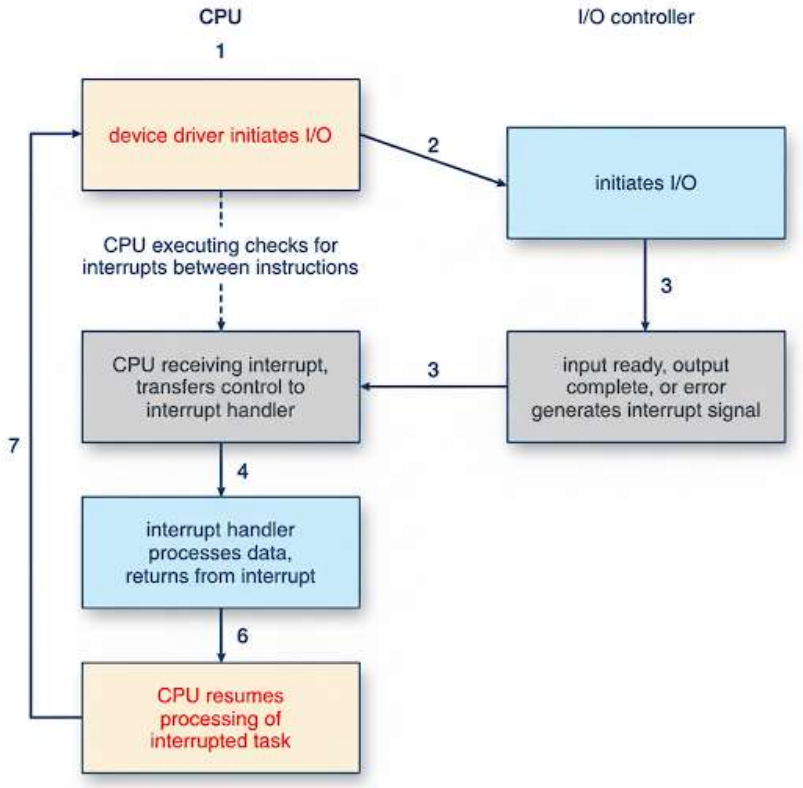
Computer-System Organization

- Hardware: A typical system has one or more CPUs and device controllers connected via a common bus
  - Each device controller has a device driver software
  - The device driver understands the device controller and provides the rest of the operating system with a uniform interface to the device
- Interrupts: The primary mechanism for hardware to signal the CPU
  - When an interrupt occurs, the CPU saves its state and transfers execution to a dedicated interrupt handler – interrupt service routine
- Storage Structure: A hierarchy of storage based on speed, size, and cost
- Volatile: Main memory (DRAM)
- Nonvolatile: Hard-disk drives (HDDs) and nonvolatile memory (NVM) devices
- Direct Memory Access (DMA): A controller that transfers data directly between a device and memory, freeing the CPU from this task

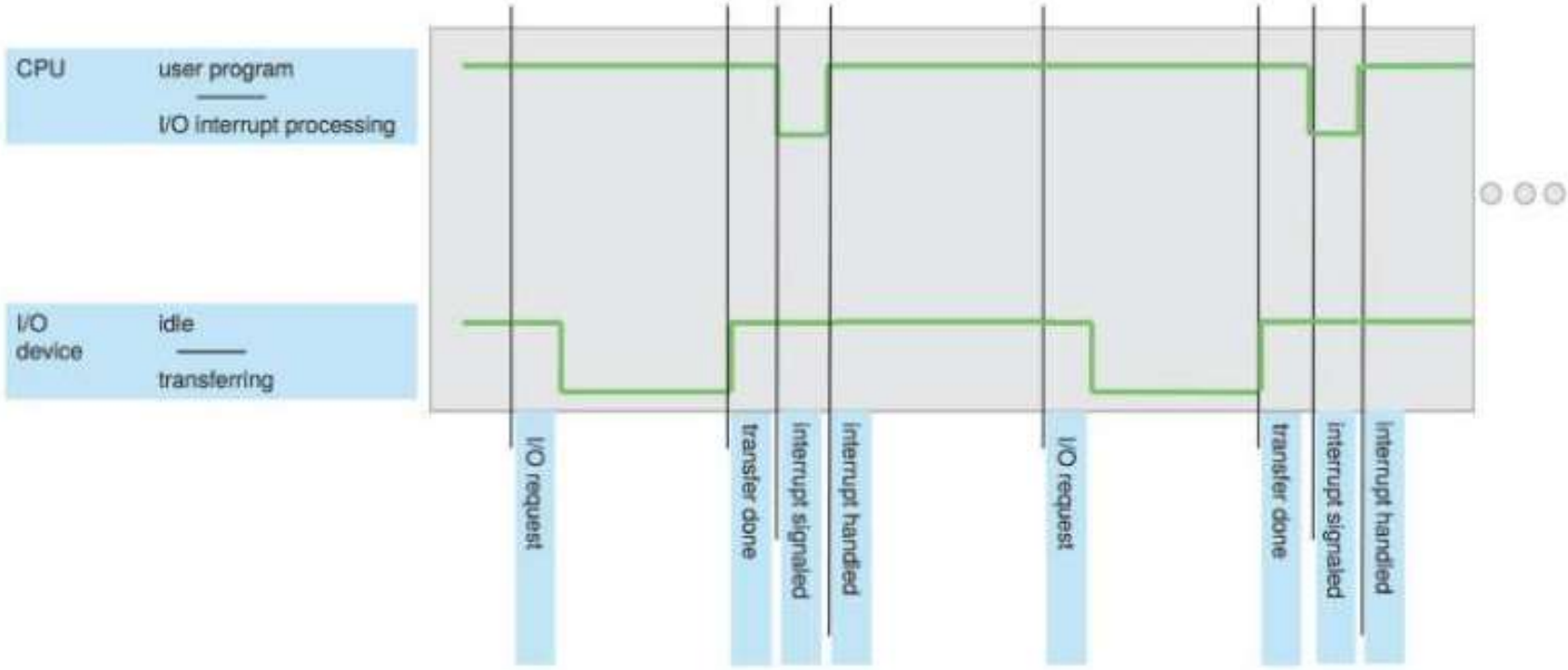


Interrupt

- Interrupt Mechanism: Hardware signals the CPU to request service
  - The CPU saves its state and transfers execution to an interrupt handler routine
  - This routine services the interrupt and then the CPU resumes its original task.
- Interrupt-Driven I/O: The primary mechanism for hardware to signal the CPU that an I/O operation is complet
  - A device controller informs the CPU by raising an interrupt
- Interrupt Vector: A table of pointers to interrupt service routines stored in low memory
  - An interrupt number is used as an index into this table to quickly find the correct handler.
- Types of Interrupts
  - Non-maskable Interrupts: Reserved for critical, unrecoverable events (e.g., memory errors)
  - Maskable Interrupts: Can be turned off by the CPU for critical instruction sequences; used by device controllers
- Interrupt Priority: A system of priority levels allows the CPU to handle high-priority interrupts before low-priority ones, ensuring urgent tasks are completed first

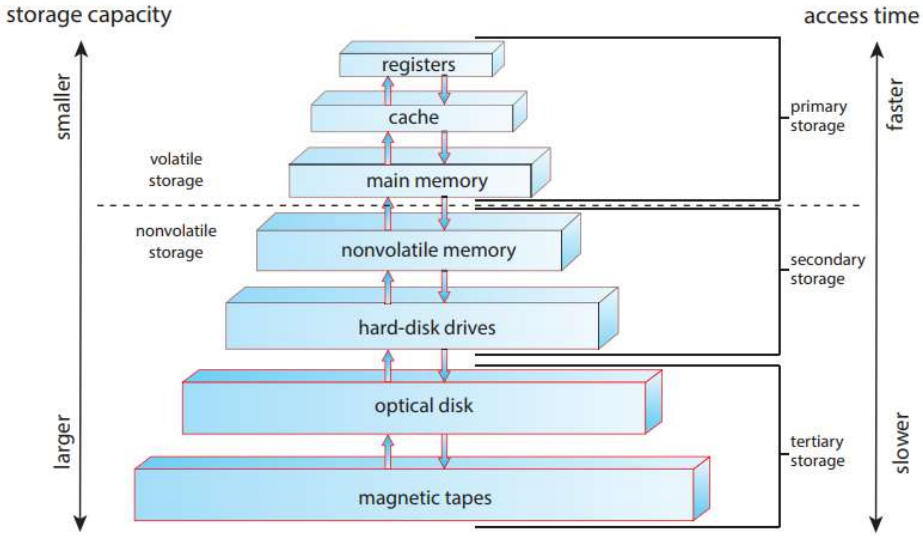


Interrupt Timeline for a Single Program



Storage

- Storage Hierarchy: Organized based on speed, cost, and volatility. The closer to the CPU, the faster and more expensive the memory
- Main Memory (RAM)
  - Volatile: Loses its contents when power is turned off
  - Directly accessible by the CPU
- Secondary Storage
  - Nonvolatile: Stores data permanently.
  - Examples include Hard-Disk Drives (HDDs) and Nonvolatile Memory (NVM) devices
- Direct Memory Access (DMA): A controller that transfers data directly between a device and memory, freeing the CPU from this task

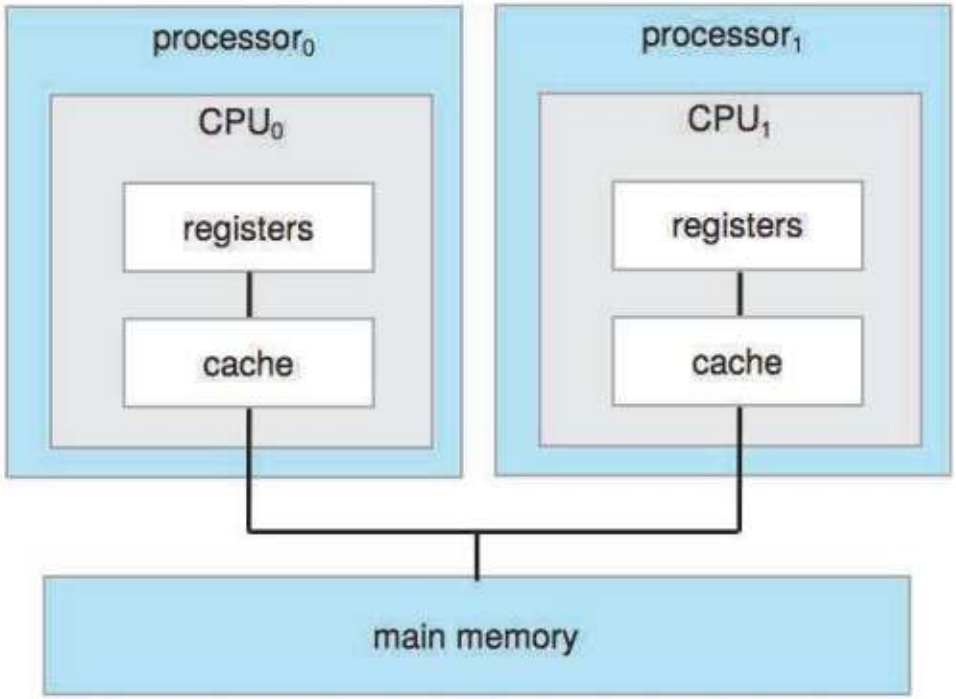


Computer-System Architecture

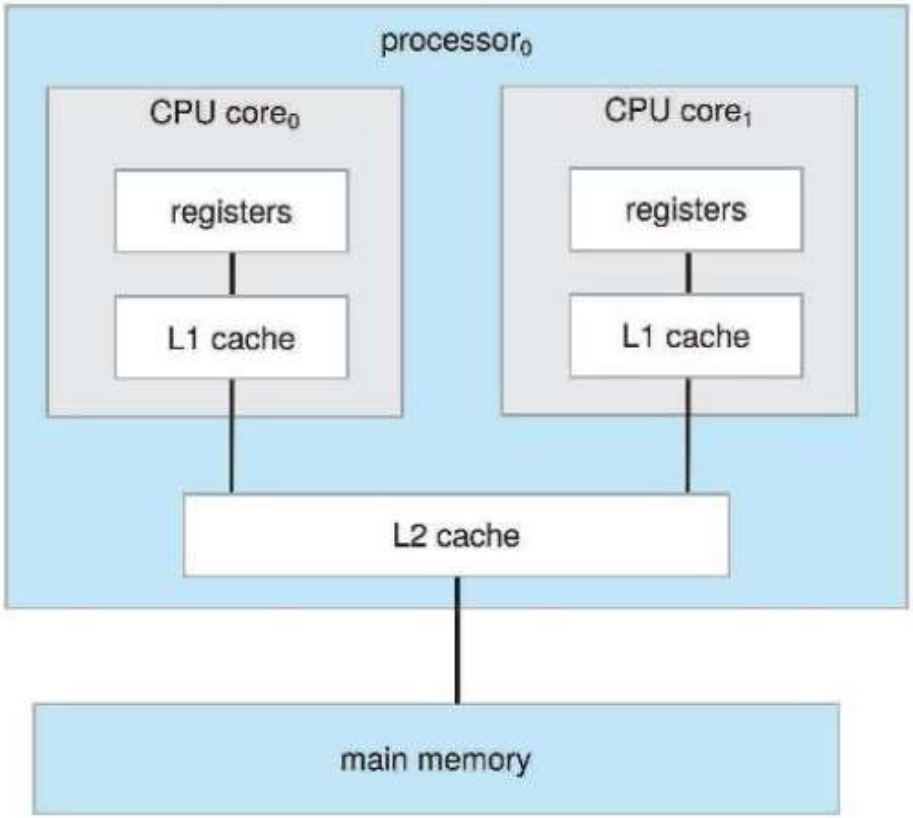
- Single-Processor Systems: A single CPU with one core, plus special-purpose processors for devices
- Multiprocessor Systems: Systems with multiple CPUs or cores that can run tasks in parallel
- Symmetric Multiprocessing (SMP): Each processor is a peer and can perform all tasks
- Multicore Systems: Multiple cores on a single chip, which is more efficient
- Clustered Systems: Multiple individual systems (nodes) linked together for high availability and high-performance computing

SMP and Multi-core Systems

SMP



Multi-core



Operating-System Operations

- Bootstrap Program: Firmware code that loads the kernel when the computer starts
- Multiprogramming & Multitasking: Techniques to keep the CPU busy by running multiple processes concurrently
  - Multitasking switches between processes so frequently that users perceive continuous interaction
- Dual-Mode Operation: Provides user mode and kernel mode to protect the OS from malicious user programs
  - The mode bit indicates the current mode
- Timer: A hardware timer interrupts the CPU after a specified period, ensuring the OS maintains control and prevents a single process from running indefinitely

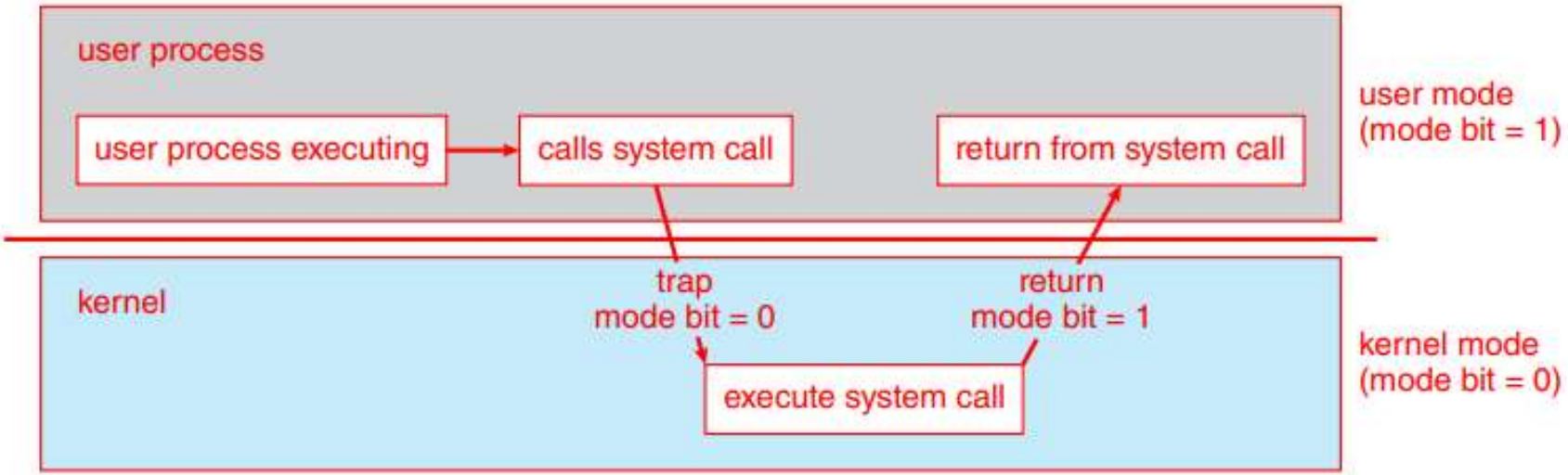
Multi-tasking Systems

- In a multitasking system, the operating system must ensure reasonable response time
- A common method for doing so is virtual memory, a technique that allows the execution of a process that is not completely in memory
- The main advantage of this scheme is that it enables users to run programs that are larger than actual physical memory
- It abstracts main memory into a large, uniform array of storage, separating logical memory as viewed by the user from physical memory
- It frees programmers from concern over memory-storage limitations

Kernel and User Mode

- Operating system and its users share the hardware and software resources of the computer system
  - Must ensure that an incorrect (or malicious) program cannot cause other programs to execute incorrectly
- Operating system must be able to distinguish between the execution of operating-system code and user-defined code
  - Approach by most computer systems is to provide hardware support that allows differentiation among various modes of execution – mode bit
- Two modes of operation
  - Kernel and User mode
- Privileged instructions: Machine instructions that only executes in kernel mode

User Mode to Kernel Mode



Resource Management

- Process Management: The OS creates, deletes, schedules, and synchronizes processes
- Memory Management: The OS tracks and allocates memory to processes, deciding what pages to move in and out of physical memory
- File-System Management: The OS provides a logical view of storage and manages file and directory structures
- Mass-Storage Management: The OS manages disk scheduling, free-space allocation, and partitioning for secondary storage
- Caching: Data is copied to a faster storage system (cache) to improve performance
  - The OS manages the movement of data in the storage hierarchy

Process Management

- Process Definition: program in execution
  - A process needs resources like CPU time, memory, and I/O devices to accomplish its task
- Process Lifecycle
  - A process is a passive entity (like a file) that becomes active when loaded into memory
- Multiple processes can be in memory and ready to run simultaneously
- The OS switches the CPU between them to maximize utilization
  - Process Control Block (PCB): A data structure that stores all information needed to manage a process, including its state, program counter, CPU registers, and scheduling information
- A process ends its execution by calling “exit()”, at which point its resources are reclaimed by the operating system

Memory Management

- Main memory is a large array of bytes that the CPU and I/O devices can quickly access
  - It is the only large storage device that the CPU can address directly
- Logical vs. Physical: For a program to be executed, it must be loaded into memory
  - As the program runs, it generates logical addresses that are mapped to physical addresses in main memory
- The operating system is responsible for the following activities in connection with memory management:
  - Keeping track of which parts of memory are currently being used and which process is using them
  - Allocating and deallocating memory space as needed
  - Deciding which processes (or parts of processes) and data to move into and out of memory

Security and Protection

- Protection: A mechanism for controlling access of programs, processes, or users to system resources
  - Mechanism must provide means to specify the controls to be imposed and to enforce the controls
  - Reliability can be improved by detecting latent errors at the interfaces between component subsystems
- Security: Defends the system against external and internal attacks, ensuring the integrity of data and resources
- User IDs & Group IDs: Used to distinguish users and groups, enabling the OS to enforce protection and security policies
  - A user sometimes needs to escalate privileges to gain extra permissions for an activity

Virtualization

- Abstracts the hardware of a single computer into multiple execution environments
- Allows multiple operating systems to run concurrently, provides isolation between environments, and facilitates system development, testing, and consolidation
  - Virtual machine
- Emulation involves simulating computer hardware in software which is typically used when the source CPU type is different from the target CPU type
- Cloud Computing
  - A major form of virtualization where resources are delivered as a service over a network

Distributed Systems

- A collection of separate, networked computers that provide a unified environment
- Advantages: Resource sharing, computational speedup, and improved reliability
- Networks
  - Local-Area Networks (LANs): Connect computers in a small geographical area (e.g., a building)
  - Wide-Area Networks (WANs): Link computers over large distances (e.g., the Internet)

Kernel Data Structures

- Lists: Used to represent collections of data values as a sequence. Can be singly linked, doubly linked, or circularly linked
- Stacks: A last-in, first-out (LIFO) data structure
- Queues: A first-in, first-out (FIFO) data structure
- Trees: Used to represent data hierarchically, such as binary search trees
  - Hash Maps: Used to associate keys with values for fast data retrieval
  - Bitmaps: A string of binary digits used to represent the status of items, such as free disk blocks

Computing Environments

- Traditional Computing: PCs connected to a network, often with local servers.
- Mobile Computing: Computing on handheld devices like smartphones and tablets, which are characterized by portability, touch-screen interfaces, and specialized hardware (e.g., GPS, accelerometers)
- Client-Server Computing: Server systems satisfy requests from client systems (e.g., file servers, compute servers)
- Peer-to-Peer Computing: All nodes are considered peers and can act as both clients and servers
- Cloud Computing: Resources are delivered as a service over a network
- Real-Time Embedded Systems: Systems with rigid time constraints, running real-time operating systems for specific tasks (e.g., car engines, industrial control systems)

Introduction